

CS4.406: Information Retrieval & Extraction

Assignment 2 Exam-Style Question Bank

A2-heavy practice set built from the Assignment 1 / Quiz-1 pattern: implementation-specific questions, quantitative reasoning, system design, evaluation, leakage, ablations, serving, and scale. No solutions are included.

Section	Focus	Approx. emphasis
A	Your actual A2 system: features, re-ranker, baseline, ablation	Very high
B	Latency, throughput, memory, cost, scaling	High
C	Evaluation, slices, bootstrap CIs, interpreting gains	High
D	Leakage, debugging, failure analysis, design trade-offs	High
E	A1 foundations carried into A2	Moderate
F	Integrated case-study / derivation questions	High

How to use: For questions marked **[Personal]**, answer from your own repository and measured experiments. State dataset/split, actual libraries, parameters, files, and measured numbers. If you did not measure something, label an estimate and show the reasoning. For quantitative questions, show equations before substituting numbers.

A. Personalised A2 implementation questions

1. [Personal] Choose one dataset (MIND or EB-NeRD) and draw the exact data flow in your submitted A2 system from an impression to the final ranked list. Name the file/module/function at each major stage.
2. [Personal] State the candidate-generator method inherited from A1, the value of K used before re-ranking, and why that K was chosen. Give one measured recall number supporting the choice.
3. [Personal] Inventory every behavioural feature used by your re-ranker. Group them into click-history, session, and article features and state the exact representation of each.
4. [Personal] For your recency-weighted history feature, write the exact decay formula you implemented, define every symbol, and state the decay hyperparameter.
5. [Personal] Show, for one real or synthetic user, how the feature vector changes when one clicked article is 5 minutes old versus 2 days old. Use your actual feature definition.
6. [Personal] Which features depend on the order of clicks inside a session? Explain how your implementation captures ordering, if at all.
7. [Personal] State whether dwell time was available in your chosen dataset and exactly how you used it, ignored it, or approximated it.
8. [Personal] How did you represent article popularity? Give the counting window, normalization (if any), and whether the value was available without future information at serving time.
9. [Personal] How did you represent freshness? Write the exact quantity you computed from publication time and impression time.
10. [Personal] Give one example where category match between a candidate article and user history would increase its score. Explain which feature value changes.
11. [Personal] Describe your behavioural-window boundary. For one impression at time t , state exactly which clicks are legal features and which are leakage.
12. [Personal] Name the re-ranker model and its actual hyperparameters: number/depth of trees or neural layers, learning rate, regularization, batch size/epochs, and early-stopping rule, as applicable.
13. [Personal] State the exact feature-to-score computation used by your re-ranker. Which parts are learned and which are fixed engineering choices?
14. [Personal] Give the exact interface between the A1 candidate generator and the A2 re-ranker: input columns/tensors, candidate count, score fields, and final ordering.
15. [Personal] Reproduce your baseline: identify the baseline implementation, its repository/file path, important parameters, and the exact data split on which you reproduced it.

- 16. [Personal]** State the one principled change you introduced beyond the baseline. Explain the hypothesis it was intended to test.
- 17. [Personal]** Present your main ablation: what component was removed or changed, what stayed fixed, and what metric difference resulted?
- 18. [Personal]** Give the paired bootstrap 95% confidence interval for your claimed gain. State the paired unit, the statistic computed per pair, and the resampling procedure.
- 19. [Personal]** Identify one result where the improved model changed ranking without changing candidate recall. Explain why this can happen in a retrieve-then-rank pipeline.
- 20. [Personal]** Identify one result where the re-ranker could not recover a relevant article because it was absent from the candidate set. What A1-side change would address this limitation?
- 21. [Personal]** State the RAM footprint of your ANN index and feature store, and explain how you measured each.
- 22. [Personal]** State p99 latency for candidate generation, p99 latency for re-ranking, and p99 end-to-end latency. Explain how many requests/examples were timed and whether warm-up was used.
- 23. [Personal]** Give your measured indexing throughput for the lexical and semantic components and explain the dominant bottleneck in each.
- 24. [Personal]** Describe the exact optimization mechanisms used in your serving/evaluation path: batching, caching, precomputed user vectors, vectorized matrix products, multiple processes, or others. Quantify at least one gain.
- 25. [Personal]** At 10x the current load, identify the first resource or stage you expect to saturate. Support the claim using your measured benchmark rather than intuition alone.
- 26. [Personal]** Compare your baseline and improved system on AUC, MRR, nDCG@5, and nDCG@10. Which metric changed the most and what property of the ranking explains that pattern?
- 27. [Personal]** Compare cold-start versus warm users. Which behavioural features should theoretically weaken for cold-start users, and what did your measurements show?
- 28. [Personal]** Compare head versus tail articles. Explain how popularity and freshness features may interact with this slice.
- 29. [Personal]** Report diversity, novelty, and coverage before and after re-ranking. Explain a case where a relevance gain could coexist with a diversity loss.
- 30. [Personal]** Identify one surprising failure case from your experiments. Trace it through candidate generation, feature construction, model score, and final rank.

B. Quantitative, derivation, and back-of-the-envelope questions

1. An impression has 150 candidates before re-ranking. Candidate generation takes 18 ms and scoring each candidate takes 0.12 ms on average in a sequential implementation. Estimate total mean request time and explain whether the scoring stage or retrieval stage dominates.
2. A re-ranker processes candidates in batches of 50. A 150-candidate impression therefore requires 3 batches. If each batch costs 2.4 ms and there is 0.8 ms fixed overhead per request, compute the idealized request latency excluding data loading.
3. Your system scores 8,000 impressions in 40 seconds using one worker. Compute throughput in impressions/s. If mean response time is 5 ms per impression, check whether the throughput is compatible with a single-worker Little's-law-style bound.
4. Suppose p99 end-to-end latency is 82 ms at 50 QPS and 146 ms at 100 QPS. What does this suggest about utilization and queueing? State at least two hypotheses that you would test.
5. An ANN index contains 160,000 article vectors of dimension 768 stored as float32. Ignoring metadata and ANN overhead, estimate raw vector memory. Repeat for float16 and state the percentage reduction.
6. A feature store has 20 features per candidate, 200 candidates per impression, and 10,000 impressions in a benchmark. Estimate the number of feature values processed if no features are cached.
7. Your lexical index has 5 million postings and each posting requires 12 bytes for document id, term frequency, and pointer metadata. Estimate raw posting storage and give one reason the real index can be larger or smaller.
8. A re-ranker improves MRR from 0.312 to 0.328 on 100,000 evaluation impressions. Compute the absolute gain and relative percentage gain.
9. A baseline nDCG@10 is 0.401 and the improved system is 0.417. Compute the absolute difference. Then explain why this difference does not by itself establish statistical significance.
10. In a paired bootstrap experiment, 2,000 bootstrap resamples produce gain estimates. The 2.5th and 97.5th percentiles are -0.001 and 0.014. Interpret the result and state whether the assignment's claimed-gain criterion is met.
11. Suppose a 95% CI for the gain in AUC is [0.004, 0.011], but the CI for nDCG@5 is [-0.002, 0.008]. What can you conclude about the claim that the improvement is consistently beneficial across both metrics?
12. At target p99 < 100 ms, one CPU instance sustains 18 QPS while a traffic router requires 2 redundant instances at all times. Ignoring network overhead, estimate the minimum number of active instances for 200 QPS if you require capacity headroom of 20%.
13. Your feature computation uses exponential decay $w = \exp(-\lambda * \Delta_t)$. For $\lambda = 0.1$ and $\Delta_t = 5$, compute w . Repeat for $\Delta_t = 20$ and compare the relative contribution.
14. A popularity count for an article is 900 in one window and 1,800 in another. If you use $\log_{1p}(\text{count})$, compute both values and explain why log scaling can be useful.
15. A model uses $K=200$ candidates and reduces to $K=50$ after a first-stage score threshold. Candidate scoring cost is linear in K . What theoretical percentage reduction in scoring work does this produce? Name the retrieval metric you must monitor to ensure the reduction is not harming recall.

C. Design choices, comparisons, and 'why' questions

1. Why is it unsafe to train behavioural features using clicks that occur after the timestamp of the impression? Give a concrete example involving popularity or recent-click features.
2. Why can a very strong re-ranker fail to improve end-to-end recommendation quality even when its local ranking loss decreases?
3. Explain the difference between candidate-generation recall and final ranking quality. Construct an example where candidate recall is high but nDCG@10 is poor.
4. Compare a GBDT re-ranker and a small neural ranker for this assignment. Discuss at least three differences in feature handling, training cost, interpretability, and serving.
5. When would you prefer a recency feature over a raw click-count feature? Give a news-recommendation scenario and explain the temporal assumption.
6. When can a freshness feature become a proxy for popularity or exposure rather than genuine relevance? Describe a possible confound.

- 7.** Why should an ablation change one principled component at a time while keeping other factors fixed? What goes wrong when data splits or hyperparameters also change?
- 8.** Why is a paired bootstrap more appropriate than an unpaired comparison when the same impressions are scored by baseline and improved systems?
- 9.** Suppose the improved system raises MRR but lowers coverage substantially. Give two plausible mechanisms that could produce this result.
- 10.** Why can head-vs-tail slicing reveal behaviour that is hidden by an overall average metric?
- 11.** Why might cold-start users benefit more from article-level features than from long click-history encoders?
- 12.** Explain why position bias is a problem when click logs are treated as direct relevance labels.
- 13.** Why should serving-time features be distinguished from offline-only features? Give two examples of information that might accidentally be available offline but not at serving time.
- 14.** What are the trade-offs of increasing K from 100 to 200 before re-ranking? Discuss recall, latency, memory traffic, and opportunity for the ranker to improve ordering.
- 15.** Why can caching user representations reduce latency but introduce consistency/staleness concerns?
- 16.** Why might an ANN index be preferable to brute-force semantic retrieval at scale, even if brute force is simpler?
- 17.** Why can increasing ANN approximation or reducing search effort change downstream ranking metrics in a two-stage system?
- 18.** Why is 'the model is better because the leaderboard score increased' insufficient as the main scientific argument in an assignment report?
- 19.** A baseline is reproduced on MIND but not EB-NeRD. List the classes of mismatch you would audit before attributing the difference to language or dataset properties.
- 20.** A temporal split gives an apparently lower score than a random split. Explain why that can be expected in rapidly changing news data.
- 21.** Your improved model uses freshness weighting. Suggest one experiment that distinguishes genuine freshness benefit from leakage.
- 22.** A feature has a large gain in an ablation, but its online value is unavailable. What should the final evaluation report say?
- 23.** How can you tell whether a latency regression comes from retrieval, feature construction, model inference, or serialization? Design a measurement plan.
- 24.** Explain why an end-to-end benchmark should report the traffic distribution or test-set characteristics rather than only a single average latency.
- 25.** Suppose two systems have identical nDCG@10 but different diversity and coverage. What does that tell you about the information lost by a relevance-only summary?

D. Debugging, leakage, statistical reasoning, and scale

1. Debugging: Your training pipeline passes the leakage test, but the validation score is implausibly high. List five concrete implementation checks, ordered from data processing to model evaluation.
2. Debugging: After adding recency-weighted history, AUC improves but MRR and nDCG@5 fall. Give three hypotheses and one experiment for each.
3. Debugging: Candidate recall@200 is 0.96, but after re-ranking the ground-truth clicked article is frequently pushed below rank 50. What does this indicate about the failure mode?
4. Debugging: p99 latency doubles after adding one new feature. Design a profiling experiment to locate the source without changing model architecture.
5. Debugging: Two team members report different metrics from the same commit. Identify the configuration, split, random-seed, preprocessing, and evaluation artefacts that must be logged.
6. Leakage: A popularity feature is computed globally over the whole dataset. Construct a situation where future clicks alter a training feature and make validation optimistic.
7. Leakage: The user's mean clicked-article embedding is computed from the complete history rather than history up to the impression. Explain exactly how the leakage changes the input representation.
8. Leakage: You use publication timestamp to compute freshness, but some records contain inconsistent time zones. Explain one way this can systematically distort the feature.
9. Failure analysis: Your system improves warm-user nDCG but hurts cold-start nDCG. What component would you inspect first, and why?
10. Failure analysis: Tail-article coverage improves but novelty falls. Give an interpretation consistent with both observations.
11. Experiment design: Propose a three-arm experiment that separates the effects of A1 candidate quality, behavioural features, and the learned re-ranker.
12. Experiment design: Design a benchmark that isolates index lookup time, feature extraction time, model inference time, and ranking/sorting time.
13. Statistical reasoning: A single random seed yields a significant gain, but three additional seeds do not. How should the result be reported?
14. Statistical reasoning: A gain is statistically significant but practically tiny. What additional evidence would you report before calling it a meaningful system improvement?
15. Scale: Your local test uses 10,000 impressions, but the deployment target is millions. List the assumptions behind extrapolating cost/QPS from the local benchmark.
16. Scale: At 10x load, which parts of a two-stage system scale linearly, which may scale sublinearly, and which can develop queueing effects? Explain with examples.

E. Assignment 1 foundations that remain relevant in Assignment 2

1. A1 carry-over: For BM25 retrieval, explain why an inverted index is needed and what information a posting must support scoring.
2. A1 carry-over: Given the same user history, explain one case where lexical retrieval can beat semantic retrieval and one case where semantic retrieval can beat lexical retrieval.
3. A1 carry-over: Why are temporal train/validation/test splits especially important for news recommendation?
4. A1 carry-over: State the formulas for precision-style recall@K, MRR, nDCG@5, and nDCG@10 as used in your evaluation, including how multiple relevant items are handled.
5. A1 carry-over: Explain why candidate-generation recall@200 can be a useful upstream diagnostic for A2 re-ranking.
6. A1 carry-over: What does an ANN index trade away compared with exact brute-force nearest-neighbour search?
7. A1 carry-over: Compare title+abstract BM25 against embedding retrieval in terms of lexical matching, paraphrase handling, index structure, and computational cost.
8. A1 carry-over: Why should article embeddings be precomputed when article content is static and the online query is a user representation?
9. A1 carry-over: If your A1 lexical and semantic retrievers each return 100 candidates, describe two ways to form the A2 candidate set before re-ranking.

- 10.** A1/A2 bridge: Suppose lexical retrieval has stronger tail recall while semantic retrieval has stronger head recall. How might you exploit this observation in A2?
- 11.** A1/A2 bridge: If a candidate is missing from both A1 retrievers, can the A2 re-ranker recover it? Explain the architectural limitation.
- 12.** A1/A2 bridge: Which A1 evaluation measurements would you retain after adding a re-ranker, and which new measurements become necessary?

F. Integrated A1 -> A2 case studies

1. Integrated case: An impression contains 120 candidates. BM25 alone puts the clicked article at rank 73; semantic retrieval puts it at rank 31; the A2 re-ranker puts it at rank 4. Explain which pipeline stages contributed what, and which single measurement would verify that A2 did not 'create' the candidate.
2. Integrated case: Baseline MRR is 0.290 and improved MRR is 0.307. The paired bootstrap CI for the gain is [0.006, 0.028]. The improved model also raises coverage from 0.34 to 0.39 but lowers novelty from 0.61 to 0.55. Write a compact interpretation suitable for the report, without claiming more than the evidence supports.
3. Integrated case: A model uses 10 recent clicks, each represented by an embedding, and averages them. Propose two alternative user encoders: one purely feature-based and one attention-based. State what failure mode each could address.
4. Integrated case: A feature vector contains click count, exponential recency, freshness, popularity, and category match. Construct a candidate pair where popularity favours article A but freshness and category match favour article B. Explain which model families can learn nonlinear interactions among these signals.
5. Integrated case: A team claims a 4% nDCG@10 improvement after changing both the re-ranker and the candidate K from 100 to 200. Explain why the experiment does not isolate the claimed architectural improvement and design a corrected experiment.
6. Integrated case: On MIND, warm-user MRR rises by 12% relative while cold-start MRR is unchanged. On EB-NeRD, warm-user gain is small but cold-start gain is positive. Give two data or feature explanations that are consistent with this pattern and suggest a test for each.
7. Integrated case: At 50 QPS, p99 is 58 ms; at 100 QPS, p99 is 72 ms; at 150 QPS, p99 is 138 ms. Sketch the qualitative latency-vs-load curve and explain what operational phenomenon is likely emerging.
8. Integrated case: Your ANN index fits in 2.5 GB at float32. A deployment budget allows 1.5 GB. Give two engineering options and state the accuracy/latency risks of each.
9. Integrated case: A future-click leakage bug increases AUC from 0.71 to 0.89. The corrected score is 0.73. Explain why the magnitude of the drop is not evidence that the corrected system is poor.
10. Integrated case: You have one hour to decide whether to spend effort improving candidate recall or the re-ranker. The current recall@200 is 0.61, and the re-ranker has already achieved strong separation among retrieved candidates. Explain what evidence you would inspect before choosing where to invest.
11. Integrated case: Two systems have the same final nDCG@10. System A uses 200 candidates and a neural re-ranker; System B uses 100 candidates and GBDT. Formulate a serving-oriented comparison using at least four measurements.
12. Integrated case: A feature store is shared by offline training and online serving. Describe a data contract that would reduce the risk that the two paths compute different definitions of the same feature.

Common examination-style patterns to practise

- Name the mechanism, then justify it using the numbers or implementation evidence you actually observed.
- State the exact boundary: time window, candidate cutoff K, train/validation split, serving-time availability, or system stage.
- Separate candidate-generation failure from ranking failure: an absent relevant item cannot be recovered by a re-ranker.
- For any claimed gain, specify baseline, improvement, ablation, evaluation unit, and paired bootstrap confidence interval.
- For serving questions, distinguish mean from p99, retrieval from re-ranking, throughput from latency, and local benchmark from 10x extrapolation.
- When comparing slices, state the population and explain the mechanism that could make the slices behave differently.

Source basis: Assignment 1, Assignment 2, and the Assignment-1-linked Quiz-1 Q2 pattern supplied in this conversation. The question bank intentionally places greater weight on A2's behavioural features, two-stage re-ranking, reproduced baseline + principled improvement, paired bootstrap significance, serving/scale analysis, and extended slicing while retaining A1 retrieval/evaluation foundations.